



Lab #3

Application Acceleration with High-Level Synthesis

National Taiwan University

2021 Fall



Purpose

- Develop application accelerator with OpenCL for Xilinx Alveo U50 PCIe FPGA (Cloud)
- A framework for future application acceleration project.



Server Login

- We have set up some Linux servers with Xilinx Alveo U50 PCIe FPGA in the lab.
- You can connect them through SSH:
 - IP: 140.112.207.155
 - port: 40100, 40200, 40400
 - username & password: TBA
- Due to the limited resources of the servers, you have to **make an appointment** to use them.
 - Students who use the server without making an appointment will get some penalties. This may cause the server out of memory and terminate the building process unexpectedly.
 - https://docs.google.com/spreadsheets/d/11rIk1Sec2F7dI1M_ZlwZ9Z25vMFSJmtitkmRVwuhnyw/edit?usp=sharing



Assignment Description

- Refer to **Workbook-Lab3** for detailed steps.
- **Skip the “Installation” part** since we have installed the tools on the servers.
- Due to some technical problem, we still could not actually run the application on the FPGAs without root. Therefore, the last part of this lab (Hardware part) will be further announced.
- Now, you should first finish the **Emulation-SW** and **Emulation-HW** parts.
- You can download the materials from NTU COOL.

Report



- Brief introduction to the overall system
- What is observed and learned
- Screen Dumps:
 - Application timeline
 - Profile summary

Submission (1/2)



- Hierarchy:
 - StudentID_lab3/
 - opt1/
 - binary_container_1.xclbin.run_summary
 - opt2/
 - binary_container_1.xclbin.run_summary
 - opt3/
 - binary_container_1.xclbin.run_summary
 - opt4/
 - binary_container_1.xclbin.run_summary
 - report.pdf
- Location of “binary_container_1.xclbin.run_summary”
 - HostProject/Emulation-HW/SystemDebugger_SystemProject_HostProject/

Submission (2/2)



- Compress all above files in a single zip file named **StudentID_lab3.zip**
- Submit to NTU COOL
- Deadline: **10/31 (Sun.) 23:59**
 - 30% off for the late submission penalty



TA Information

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 - <https://meet.google.com/tfx-idxz-byw>